PRATIK SARODE

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UK WORK ELIGIBILITY | UK GRADUATE ROUTE VISA | FULL RIGHT TO WORK UNTIL JAN 2028

SUMMARY

AI Engineer with **2+ years** of experience building low-latency, multi-agent systems and advanced RAG pipelines. Backed by an MSc in Applied AI, I use LangChain, LangGraph and CrewAI to design production-ready, cloud-based LLM applications. Adept at driving complex automation projects end to end, with a strong track record of optimizing system performance and engineering robust hallucination guardrails.

TECHNICAL SKILLS

Languages: Python, Next.js, JavaScript/TypeScript

AI/ML: Machine Learning, Deep Learning, NLP, GenAI, LLMs, RAG, Prompt Engineering, AI Agents, LangGraph, MCP, Agentic AI, Evaluation, Observability (Langfuse), State management

Frameworks: TensorFlow, PyTorch, scikit-learn, LangChain, LangGraph, Hugging Face, FastAPI, Streamlit, Gradio

Backend: FastAPI, REST APIs, API Integration, Database Models, CRUD Operations, Authentication

Data/Vector Search: Pandas, NumPy, FAISS, Qdrant, PostgreSQL, Embeddings, Semantic Search

Tools and Cloud: Git, GitHub, Docker, MLflow, Jupyter Notebook, Technical Documentation, GCP (Cloud Run, GKE), Vertex AI

EXPERIENCE

AI Engineer | Gamaka Intellectual Machines *Oct 2024 - Present*

- Core engineer on real-world Generative AI and Agentic AI applications, including a Multi-Agent Autonomous Medical Insurance Claim Validation Engine.
- Architected RAG pipelines with LangChain and Langflow to optimize retrieval, improve LLM accuracy and ground enterprise responses in proprietary data.
- Orchestrated multi-agent workflows in CrewAI, using the Model Context Protocol (MCP) to maintain state and context across autonomous agents.
- Engineered prompts and integrated the OpenAI API to build conversational AI interfaces for complex financial use cases, and deployed production-ready solutions that reduced manual validation time.

KEY PROJECTS

Autonomous Medical Insurance Claim Validation Engine | Python, LangGraph, FastAPI, PostgreSQL

- **Architecture:** multi-agent clinical validation engine on LangGraph, Pydantic v2 and async Python, running six domain-specific agents in parallel after dependency-aware ICD coding, with typed outputs, checkpointing, fault tolerance and human-in-the-loop medical review.
- **Hybrid RAG:** ICD-10 coding with BM25 sparse retrieval, embeddings, Reciprocal Rank Fusion, FAISS/pgvector and constrained LLM selection, reducing unsupported code generation across clinical shorthand and paraphrased diagnoses.
- **Guardrails:** mandatory verbatim quote verification against source documents before any finding is accepted (exact match is the production gate; RapidFuzz is for debugging only). Pydantic validation gates return ABSTAIN for ambiguous, high-risk or low-evidence cases and route them to medical officers.
- **Governance and serving:** bitemporal PostgreSQL knowledge base with Git-backed governance, CI validation and medical-advisor review for audit and reproducibility. Async FastAPI layer with a Streamlit review dashboard (evidence inspection, approve/query/deny), deployed on GCP with Cloud Run for APIs and GKE for orchestration workers.
- **Evaluation and efficiency:** Langfuse, pytest, property-based tests and edge-case validation capture prompts, evidence, latency and token usage for debugging and auditing. Semantic and prompt caching, retrieval tuning and parallel execution target fewer redundant LLM calls and lower validation latency.

Agentic AI System for Fraud and AML Risk Triage | Python, LangGraph, scikit-learn, FAISS

- **ML tier:** trained and tuned scikit-learn anomaly-detection and graph-based network-analysis models on transaction data to score fraud/AML risk and pre-screen alerts before any LLM involvement.
- **Triage pipeline:** LangGraph multi-agent pipeline routes only ML-flagged high-risk alerts to LLM review, resolving most alerts in the ML/rules tier and capping LLM usage at two calls per escalated case.
- **SAR drafting:** RAG-grounded Suspicious Activity Report engine that cites every factual claim to a source text span, with a secondary LLM independently verifying claims before any statement reaches an analyst.
- **Deployment and audit:** containerised services on GKE with the ML tier and LangGraph pipeline running independently, and full audit-trail provenance across both tiers aligned with BFSI model-risk documentation standards.

EDUCATION

MSc Applied Artificial Intelligence and Data Analytics (First Class) | *University of Bradford, UK* *Sept 2024 - Sept 2025*

BE Electronics and Telecommunications (Distinction), GPA 7.76 | *AISSMS IOIT, Pune* *Jan 2021 - June 2024*

CERTIFICATIONS AND ACHIEVEMENTS

Certifications: Oracle Cloud Infrastructure Gen AI Professional | Kaggle Python Coder | AI and Agile (Scrum Alliance)

Achievements: Open-source contributor to ElizaOS DeFi Oracle Agent plugin (Nosana x ElizaOS Challenge) | Student Representative, University of Bradford | Top rank in Excellidia's AI Leadership program (50+ participants)